AlzGraph: A Knowledge Graph and Benchmark for Evidence-Grounded Alzheimer's Disease Reasoning

Anonymous Author(s)
Affiliation
Address
email

Abstract

Reasoning about Alzheimer's disease (AD) requires integrating evidence across genetics, biomarkers, disease progression, treatment, and clinical outcomes. Yet it remains difficult to evaluate whether knowledge-augmented language models can reliably perform this kind of evidence-intensive reasoning. We present ALZGRAPH, an open framework for studying knowledge-augmented reasoning in AD. The framework consists of ALZKG, a five-layer knowledge graph covering genes, biomarkers, disease stages, treatments, and outcomes; ALZBENCH, a benchmark spanning clinical decision making, report generation, biomarker-driven precision medicine, treatment recommendation, and research planning; and a Graph-RAG pipeline for retrieving multi-hop evidence from the graph and associated literature. ALZKG is constructed from public ontologies, clinical guidelines, and 7,150 full-text PubMed/PMC papers, yielding 1,224 entities and 5,065 cross-layer relations. We evaluate the framework at both the knowledge and model levels. An audit of graph relations reveals substantial extraction errors, highlighting the difficulty of constructing reliable clinical knowledge graphs from literature at scale. Retrieval experiments further show that graph-based retrieval does not consistently outperform a simple lexical retrieval baseline, indicating that multi-hop graph structure alone is not sufficient to guarantee better evidence retrieval. On downstream tasks, a deterministic KG-only baseline performs above chance on several tasks, while comparisons between models with and without Graph-RAG show mixed effects across language models. These results demonstrate both the promise and the current limitations of graph-based knowledge augmentation for AD reasoning. ALZGRAPH provides an open and reproducible testbed for systematically measuring these effects and for developing more reliable knowledge-grounded clinical reasoning systems.

1 Introduction

Alzheimer's disease (AD) is the leading cause of dementia and a growing public health challenge, affecting tens of millions of people worldwide [20]. AD is increasingly demanding as a setting for clinical AI for two reasons. First, it is now defined biologically through the ATN framework, in which amyloid, tau, and neurodegeneration biomarkers measured in CSF, plasma, and neuroimaging define a continuum from preclinical AD through mild cognitive impairment (MCI) to AD dementia [1, 15, 21, 25]. Second, the arrival of amyloid-targeting antibodies such as lecanemab and donanemab has made treatment decisions genuinely multi-factorial: eligibility depends on biomarker-confirmed amyloid and disease stage, while safety depends on genotype—*APOE* ε4 homozygotes, for instance, face substantially higher risk of amyloid-related imaging abnormalities (ARIA) [10, 23, 26].

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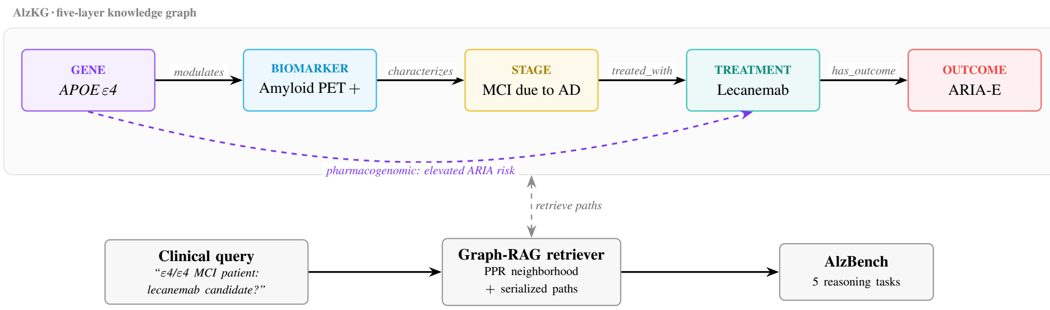

Figure 1: Overview of ALZGRAPH. ALZKG organizes knowledge into five clinical layers: genes, biomarkers, stages, treatments, and outcomes, connected by typed relations. A representative multi-hop reasoning path is highlighted. Given a clinical query, the Graph-RAG retriever identifies a personalized PageRank neighborhood and serializes relevant reasoning paths, including genotype-dependent links to ARIA risk. The retrieved evidence is evaluated across the five tasks in ALZBENCH.

Answering AD clinical questions therefore requires multi-hop reasoning: a risk gene links to a biomarker profile, the profile defines a stage, the stage gates a therapy, and the therapy is constrained by genotype-dependent safety. Knowledge graphs are a natural substrate for this kind of reasoning, but existing biomedical KGs are largely disease-agnostic or built for semantic annotation [7, 9], and none organizes the gene–biomarker–stage–treatment–outcome structure AD reasoning demands. Large language models, meanwhile, show strong medical question-answering ability [24] and can be grounded in external evidence through retrieval-augmented generation [11, 19], but most medical benchmarks test isolated, short-form questions rather than the integrative, safety-critical reasoning AD care requires. A gap remains for tooling that tests whether generalist models can curate and reason over AD evidence the way this multi-hop structure demands.

Present work. We introduce ALZGRAPH, a framework with two components (Figure 1). **ALZKG** is an Alzheimer's knowledge graph built through a structured evidence-to-graph pipeline: a five-layer schema (*genes, biomarkers, stages, treatments, outcomes*) grounded in NIA-AA, OMIM, the GWAS Catalog, HGNC, MeSH, HPO, ChEBI, UMLS, and AAN/Alzheimer's Association guidelines, populated by mining 7,150 PMC full-text papers with sentence-level relation extraction. **ALZBENCH** is a five-task benchmark built around this graph, with a Graph-RAG retriever that serializes literature-weighted, multi-hop reasoning paths for language models. Rather than assume a mined knowledge graph is useful once built, we evaluate the pipeline itself: we manually audit extracted relations against their source text, and we test whether graph-based retrieval and downstream reasoning actually benefit from the graph's structure relative to simpler baselines. These experiments surface both useful structure and real limitations, including substantial extraction noise and cases where a flat retrieval baseline matches or beats Graph-RAG. In summary, we contribute (1) ALZKG, a five-layer AD knowledge graph with literature-grounded, multi-hop relations; (2) ALZBENCH, a five-task benchmark and Graph-RAG retriever for evaluating any LLM; and (3) an empirical audit of both the graph's extraction quality and its downstream usefulness, providing an open, reproducible testbed for studying when structured biomedical knowledge actually helps LLM-based clinical reasoning.

2 AlzKG: Alzheimer's Knowledge Graph

2.1 Problem Formulation

We represent ALZKG as a multi-relational heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with each entity in $\mathcal{V}$ assigned to one of five clinical layers—*genes, biomarkers, stages, treatments,* or *outcomes*—and each fact a typed triplet $\langle h, r, t \rangle$. We focus on cross-layer relations, since these are what support the multi-hop reasoning diagnosis and treatment selection require. Each relation carries an evidence weight: a curated tier (1–3, emerging to guideline-level) for seed relations from guidelines and ontologies, or a supporting-paper count for relations mined from the literature (§2.3).

2.2 Data Collection and Processing

ALZKG is built from Alzheimer's disease literature retrieved from PubMed/PMC via NCBI E-utilities, using five AD-focused queries (genetics, amyloid/tau biomarkers, imaging and staging, treatments, and clinical outcomes) restricted to 2010–2026. After deduplication, we retrieve full text for every article with an open-access PMC record, yielding **7,150 full-text papers**. Each article is segmented into sentences, and we keep sentences containing at least two recognized ALZKG entities as candidate evidence units for relation extraction—**320,299** sentences in the released corpus, which reflects the corrected pipeline of §2.5 (in particular, a fix to a data-completeness bug that had let some papers silently fall back to abstract-only content).

The entity vocabulary is grounded in established resources: NIA-AA [15] for biomarker/stage terminology, OMIM/ADSP/GWAS Catalog [3, 18] for a curated AD-gene seed set, MeSH/HPO [17] for phenotype terms, ChEBI [14] for drug identifiers, AAN/Alzheimer's Association criteria [10] for treatment concepts, and UMLS [5] for cross-ontology normalization. Gene recognition deliberately extends beyond this curated seed to the full HGNC catalog ($\sim$19,000 approved symbols, matched case-sensitively so common words are not mistaken for genes), so that extraction—not vocabulary coverage—determines whether an emerging AD gene enters the graph (§2.5 evaluates the resulting precision trade-off). Surface mentions are normalized to canonical entities via a curated synonym map (e.g. "Leqembi"$\rightarrow$*Lecanemab*, "Aβ42"$\rightarrow$*Amyloid-β 42*), matching longer phrases before shorter substrings to reduce overlapping matches.

2.3 Knowledge Graph Construction

Entities are recognized per sentence with this same dictionary-based lexicon; representative concepts per layer include *APP*/*PSEN1*/*APOE*/*TREM2* (genes), CSF/plasma Aβ42, p-tau181/217, amyloid/tau PET, MMSE/MoCA (biomarkers), preclinical AD through severe dementia (stages), lecanemab/donanemab (treatments), and ARIA/cognitive decline (outcomes). Rather than treat every entity co-occurrence as a candidate edge, we impose a typed schema in which each pair of layers admits exactly one relation and direction reflecting the underlying clinical assertion—e.g. a gene may be a risk factor for a stage or modulate a biomarker (all ten relation types and their layer pairs are in Table 2). An edge additionally requires textual support: for a sentence containing an entity pair whose layers match a schema relation, we further require a relation-specific trigger phrase (e.g. "associated with," "increases the risk of," "treated with"); co-occurrence alone is never sufficient. Each accepted triplet retains its supporting sentence(s) and source paper(s) as provenance, and duplicate triplets from different sentences or papers are merged. We keep a relation only if at least three distinct papers support it, to suppress isolated or spurious mentions, and weight it by that paper count; the small set of ontology/guideline-derived seed relations instead carry a curated evidence tier. The released graph uses only this deterministic, trigger-based procedure—an optional LLM-based extraction module exists but was not used for any result in this paper.

2.4 Graph Mapping and AlzKG Statistics

Recognized mentions are normalized to canonical entities, relations are assigned according to the layer-pair schema above, and duplicate triplets are merged while retaining their full set of supporting PMIDs. The resulting graph contains **1,224 entities** and **5,065 triplets** across all 10 relation types, mined from 7,150 full-text papers (Table 1); the per-relation breakdown is in Table 2. Supporting paper counts are highly skewed (median 5, mean 17.3, maximum 2,735), so a small number of well-established associations—such as $\langle$APOE, risk_gene_for, AD$\rangle$ with 1,799 papers—account for a disproportionate share of the graph's literature support. The highest-degree entities are mostly clinically central (Alzheimer's disease, dementia, cognitive decline, APOE), with one exception: *nitric oxide* ranks above APOE despite being a generic compound rather than an AD-specific concept. This is not an entity-matching error—nitric oxide is matched by its own literal name—but a symptom of the treatment vocabulary being drawn from ChEBI's general chemical hierarchy rather than curated for AD relevance, related to the precision issue we examine next.

2.5 Extraction Precision: A Manual Audit

The statistics above describe what the pipeline extracted, not whether the extractions are correct. To assess this, we manually judged random samples of triplets against their own grounding sentence—

Table 1: **Mined AlzKG statistics** (computed directly from the literature-mined graph, after the fixes of §2.5; 7,150 PMC full-text papers, 320,299 candidate sentences, minimum 3 supporting papers per edge). All 5,065 triplets are cross-layer; the maximum edge paper count is 2,735.

Graph		**Entities per layer**	
Full-text papers	7,150	Gene	723
Entities	1,224	Biomarker	72
Triplets	5,065	Stage	42
Relation types	10	Treatment	244
Median paper count	5	Outcome	143

Table 2: **AlzKG relation types, their layer pair, and mined triplet counts.** Each relation corresponds to exactly one cross-layer pair; counts sum to 5,065.

Relation type	**Layer pair**	**# triplets**
risk_gene_for	gene → stage	953
has_outcome	treatment → outcome	740
modulates_biomarker	gene → biomarker	700
associated_outcome	stage → outcome	596
gene_associated_outcome	gene → outcome	540
pharmacogenomic_consideration	gene → treatment	422
treated_with	stage → treatment	357
predicts_outcome	biomarker → outcome	290
characterized_by_biomarker	stage → biomarker	277
biomarker_guides_treatment	biomarker → treatment	190

the actual corpus sentence used to emit each edge (Appendix B.5 gives the full protocol). An initial audit of 30 triplets found only 13% fully correct and 60% unsupported, most often due to short-symbol entity collisions (e.g. the gene *NOS1* matching the clinical abbreviation "NOS") or lost relation direction. Tracing these cases surfaced two software bugs in the entity-recognition pipeline—an alias-merging step that could silently conflate unrelated concepts sharing a short string, and an order-dependent conflict-resolution step that could strip a gene of its own symbol—together with a separate data-completeness issue in which 13% of source papers had silently fetched abstract-only content. We fixed all three and rebuilt the graph.

A follow-up audit, pooled across two independent 30-triplet samples after a single draw proved too noisy to trust on its own, found precision improved only modestly: 21.7% fully correct and 55.0% unsupported (correct-or-partial rising from 40% to 45%). The bugs we fixed accounted for a minority of the errors; most of the remaining share comes from structurally different failure modes—entities merely co-listed in a panel or glossary being read as an asserted relation, laboratory techniques mislabeled as biomarkers, off-topic disease mentions, and relations that lose directionality (a drug that reduces an outcome stored identically to one that causes it). We report this still-imperfect result, rather than only the improved number, because both the retrieval analysis and the KG-only baseline below consume this same graph and inherit its precision (Appendix H discusses concrete remediation directions).

3 AlzBench

3.1 Problem Formulation

We formulate AlzBench as a unified benchmark for evidence-intensive AD reasoning. Each task is $\hat{y}_i = f_{\text{LM}}(x_i, \mathcal{E}_i, \mathcal{C}_i)$, where x_i is the task input (a clinical case, patient panel, or scientific document), $\mathcal{E}_i$ is the evidence retrieved from AlzKG via Graph-RAG, $\mathcal{C}_i$ is optional task-specific context (candidate options or metadata), and $\hat{y}_i$ is evaluated against the gold output y_i^*.

3.2 Task Design

T1: Clinical Decision Accuracy (CDA). Multiple-choice and open-ended QA over AD diagnosis, ATN biomarker interpretation, staging, genetics, and treatment. Example: *"A 71-year-old with amyloid-positive PET and mild AD dementia is APOE $\varepsilon 4/\varepsilon 4$; which adverse event is most increased if lecanemab is started?"* (answer: ARIA).

T2: Clinical Report Generation (CRG). Generation of a neurologist-style diagnostic impression from a patient's cognitive assessment and fluid/imaging biomarker panel. Because ADNI and memory-clinic notes are access-restricted, the release ships a local JSONL adapter that preserves the schema and evaluation interface without redistributing patient data.

T3: Biomarker-Driven Precision Medicine (BPM). Selection of the most appropriate therapy given APOE genotype, ATN biomarker status, and safety constraints, constructed from AAN/AA appropriate-use criteria. Example: an amyloid-positive early-AD patient on chronic anticoagulation should *avoid* anti-amyloid antibodies (ARIA-hemorrhage risk), whereas an $\varepsilon 3/\varepsilon 4$ patient without contraindications is an appropriate candidate with MRI monitoring.

T4: Treatment Recommendation (TR). Guideline-consistent dementia therapy under patient-specific constraints, built from a dementia-focused subset of MedQA-USMLE [16] with treatment-safety and KG-evidence-coverage metrics.

T5: Deep Research Planning (DRP). Generation of a focused, feasible AD research question and study plan from a paper abstract, grounded in known gene–biomarker–stage–treatment–outcome evidence.

3.3 Evaluation and Metrics

We evaluate with task accuracy and reasoning-oriented metrics. For CDA we report Top-1 accuracy (MCQ) and ROUGE-L/BLEU-1/Token-F1 (open-ended). For CRG we use ROUGE-L/Token-F1. For BPM and TR we report Top-1 accuracy and a drug-safety score (1.0 iff the selection avoids every contraindicated agent, e.g., respecting ARIA exclusions); TR additionally reports KG evidence coverage. For DRP we use ROUGE-L/Token-F1. All results in this paper use only these lexical metrics: an LLM-as-judge dimension (for open-ended CDA and CRG alignment, and DRP's scientific-validity/feasibility) is part of the task design but is not implemented in the released metrics code and was not used for any number reported here (Appendix H).

4 Experiments

4.1 Setup

The Graph-RAG retriever seeds a personalized-PageRank walk from query-mentioned entities (restart probability 0.15), keeps the highest-scoring 30-node neighborhood, and serializes up to 12 reasoning paths of depth up to four hops (Appendix D gives the full implementation and hyperparameter sweep). The evaluation code targets any model reachable through an OpenAI-compatible chat-completions interface, hosted or local; §4.4 reports results for a frontier model evaluated directly (Claude, Sonnet 5) and three open-weight models run locally (Llama-3.2-3B, Gemma3-4B, Qwen3-8B). We had no API budget for closed-source models beyond Claude, so we do not populate GPT-4o/Gemini-style rows with placeholder numbers; the three open-weight rows, none of which authored any benchmark item, are this paper's uncontaminated capability signal. Prompts use chain-of-thought reasoning, a clinical-specialist role tailored to each task, and the serialized ALZKG paths as context under Graph-RAG (Appendix F reproduces every prompt verbatim). With no budget for third-party retrieval-augmented systems, our baselines are three deterministic references instead: flat BM25 retrieval over the same corpus (§4.2), a model-free *KG-only* MCQ baseline (§4.3), and each evaluated model's own No-RAG condition.

Table 3: **Intrinsic Graph-RAG retrieval analysis on ALZKG** (30 probe queries; computed directly from the literature-mined graph). Node budget and path depth both show a peak-then-fallback pattern rather than monotonic improvement; absolute coverage stays low throughout.

Sweep	Setting	Cand. paths	Gold coverage	Latency (ms)
Subgraph nodes (d=4)	10	44.8	0.067	25.36
	20	**539.2**	**0.100**	**27.32**
	30	2,044.5	0.067	34.41
	40	5,289.2	0.067	52.04
	50	9,317.7	0.067	74.79
Path depth (n=30)	1	48.0	0.100	26.23
	2	**350.3**	**0.167**	**27.11**
	3	1,196.4	0.067	30.70
	4	2,044.5	0.067	34.94

4.2 AlzKG and Retrieval Analysis

We first evaluate retrieval quality directly, with no language model involved. Table 3 sweeps Graph-RAG's neighborhood size and path depth over 30 AlzBench probe queries, measuring *gold-evidence coverage* (the fraction of probes whose gold-answer concept is recovered in the retrieved paths) and latency. Coverage does not grow monotonically with either budget: it peaks at a compact 20-node, two-hop neighborhood (0.10–0.17) and falls back with wider or deeper retrieval, even as the number of candidate paths grows by two orders of magnitude and latency rises accordingly. This suggests that the AD reasoning paths our probes require are shallow rather than benefiting from deeper traversal, though the probe set is small enough that we treat this as suggestive rather than conclusive. We keep a wider 30-node, four-hop default in the released harness to preserve fuller reasoning paths for the language model, even though it does not maximize this coverage metric.

Comparison against a non-graph baseline. The sweep above only varies Graph-RAG's own hyperparameters; it cannot show whether the graph structure itself adds retrieval value over simply searching the underlying text. We therefore index the same candidate sentences with BM25 (no graph, no entity typing) and evaluate it on the same 30 probes with the identical coverage criterion. BM25 reaches **0.567** coverage—more than three times Graph-RAG's best setting—at a higher but still sub-second latency. The comparison is not perfectly matched: BM25 returns raw sentence text, in which a gold phrase is more likely to appear verbatim, while Graph-RAG returns compressed entity-relation paths that discard the original sentence context, which structurally favors passage retrieval on this metric. Even allowing for that asymmetry, the gap is large enough that we do not yet have evidence that AlzKG's structure improves retrieval over flat lexical search on this corpus (Appendix H discusses what would be needed to make the comparison fairer).

4.3 KG-only Baseline (Measured)

Beyond intrinsic retrieval quality, we report a fully deterministic *KG-only* baseline that answers AlzBench multiple-choice items using nothing but ALZKG evidence: it seeds the same personalized PageRank walk from entities recognized in the question stem, and selects the option whose entities receive the most evidence mass (Appendix D covers the implementation, including a reproducibility fix for a retriever tie-breaking bug). Table 4 reports results on the expanded item sets. On the hand-authored T1 and T3 items, the baseline reaches 0.50 and 0.30 accuracy, both above the 0.25 random rate. T4—general-medicine USMLE items filtered for dementia relevance rather than authored around ALZKG's vocabulary—is more revealing: on the original graph it sat at exactly the random rate, with grounded accuracy actually falling *below* chance, suggesting that graph connectivity alone does not guarantee useful evidence. On the corrected graph (§2.5) it rises to 0.375, with grounded accuracy climbing to 0.42. Because T4's items and the KG-only method are unchanged between these two measurements, this is one of the clearest model-facing signals that the extraction fixes improved the graph's usable evidence, though with only 16 items the comparison should not be over-read.

Table 4: **KG-only baseline (no LLM; deterministic, exactly reproducible), fully corrected graph (§2.5).** Multiple-choice accuracy using only ALZKG PPR evidence. *Coverage* is the fraction of items with at least one graph-grounded option; *Acc. (grounded)* restricts accuracy to those items. All three tasks now exceed the 0.25 random-choice rate; T4 (external MedQA-USMLE items, not authored around ALZKG's vocabulary) was previously at or below chance (0.25 / 0.23) on the original graph.

Task	Items	Accuracy	Random	Coverage	Acc. (grounded)
T1 CDA (MCQ)	20	**0.50**	0.25	0.80	0.56
T3 BPM (MCQ)	10	**0.30**	0.25	1.00	0.30
T4 TR (MCQ, MedQA-USMLE)	16	**0.375**	0.25	0.75	0.42

4.4 No-RAG versus Graph-RAG: A Model Comparison

For each task we compare a language model's performance without retrieval (No-RAG) against the same model augmented with Graph-RAG evidence, holding the prompt, decoding temperature, and metrics fixed across both conditions. Table 5 reports this comparison for a frontier model evaluated directly (Claude, Sonnet 5) and three open-weight models evaluated through a local deployment (Llama-3.2-3B, Gemma3-4B, Qwen3-8B). Claude reaches ceiling accuracy in every condition; because the same model instance authored the T1 and T3 items, this largely reflects well-covered parametric knowledge rather than retrieval quality, though T4's externally-sourced items reaching ceiling too is a more meaningful, if still same-model-family, data point. On T1's open-ended QA (not shown in the table), Claude obtained identical scores in both conditions.

The three open-weight models, none of which authored any item, are genuinely sub-ceiling, and Graph-RAG's effect on them is mixed rather than uniformly helpful or harmful. It costs a few points of T1 accuracy for Llama-3.2-3B while leaving Gemma3-4B and Qwen3-8B essentially unaffected. On T3 it lowers Qwen3-8B's accuracy while raising its safety score, leaves Gemma3-4B's accuracy and safety unchanged, and leaves Llama-3.2-3B's accuracy flat with a safety cost. On T4 it costs accuracy for all three models, most for Qwen3-8B despite it being the strongest model of the three under No-RAG. Qwen3's internal chain-of-thought trace also surfaced a scoring bug in our answer-extraction code, which we found and fixed (Appendix H).

We read this absence of a consistent Graph-RAG direction across models and metrics as an informative null result rather than one to filter out: on this graph and this item pool, Graph-RAG does not yet provide a reliable net benefit for open-weight models, even though the retrieved evidence is genuinely used when models reference it (§4.5). These numbers, like the KG-only baseline above, also shifted as we corrected the underlying graph; Appendix H reports the earlier measurements and discusses what this instability implies for drawing conclusions from any single such comparison.

4.5 Clinical Report Generation and Deep Research Planning

Two tasks fall short of a full benchmark comparison, and we report them as such rather than dress them up as leaderboard rows. Because ADNI and memory-clinic notes are access-restricted, T2 ships with a single synthetic patient case; we ran it through all three open-weight models in both conditions only to confirm the task runner and metrics behave correctly, not as a benchmark claim, and draw no conclusion from an n=1 comparison (Appendix H). T5 has no gold research plan to score against, so we treat it as a qualitative case study: on two real, randomly sampled corpus abstracts, Graph-RAG plans explicitly cross-referenced retrieved AlzKG evidence—for example, grounding a follow-up question in a specific, well-supported BACE1–AD edge—where No-RAG plans reasoned from the abstract alone, and both conditions produced feasible, methodologically sound designs.

5 Broader Impact

ALZGRAPH aims to make it easier to evaluate whether AI systems reason safely about Alzheimer's care, which we expect to be net positive if it raises the safety bar models are held to before clinical deployment. The main risk is over-trust: our own audit shows that most sampled ALZKG edges

Table 5: **No-RAG versus Graph-RAG accuracy across four language models.** Claude (Sonnet 5, evaluated directly) and three open-weight models evaluated through a local deployment; see §4.4 for methodology and caveats. Open-weight Graph-RAG columns are on the final, fully corrected graph (§2.5); Claude (direct) is on the original graph, both columns.

	T1 CDA (Acc)		**T3 BPM (Acc / Safety)**		**T4 TR (Acc / KGEC)**	
Model	No-RAG	Graph-RAG	No-RAG	Graph-RAG	No-RAG	Graph-RAG
Claude (direct)[‡§]	**1.00**	**1.00**	**1.00 / 1.00**	**1.00 / 1.00**	**1.00 / 0.00**[†]	**1.00 / 0.02**[†]
Qwen3-8B (local)	1.00	1.00	0.30 / 0.80	0.20 / 0.90	0.69 / 0.00	0.50 / 0.04
Llama-3.2-3B (local)	0.95	0.90	0.30 / 0.70	0.30 / 0.60	0.38 / 0.00	0.25 / 0.06
Gemma3-4B (local)	0.90	0.90	0.40 / 0.90	0.40 / 0.90	0.25 / 0.00	0.19 / 0.16

[‡]T1/T3 items were authored by the same model instance evaluated in this row; its accuracy there is confounded by self-authorship and should not be read as a capability result (Appendix H). [†]This row's T4 columns are computed on 14 of the 16 released items (2 excluded, §4.4); every other row uses all 16. [§]Measured on the original graph (§2.5) and not rerun after either subsequent fix; given this row already sits at ceiling accuracy in every condition (attributed above to self-authorship / same-model-family items rather than retrieval quality), we judged a full manual re-answer unlikely to change the reported numbers and prioritized rerunning the three open-weight rows, which are not ceiling-bound and are this table's more informative comparison.

are not supported by their grounding sentence, so surfacing individual graph edges to a clinician or patient as verified facts—rather than as unverified literature-mining output, which is what they are—could propagate a specific wrong claim, such as a treatment marked as indicated when it is not, or a safety-relevant relation with its direction silently lost. A high AlzBench score reflects performance under our specific item pools and metrics, not a certification of clinical safety or of the graph's factual accuracy (Appendix H). We do not identify a dual-use pathway specific to this release beyond those already inherent to publishing any biomedical knowledge graph or evaluation harness.

6 Conclusion and Discussion

AlzKG and AlzBench together provide a knowledge-grounded evaluation framework for Alzheimer's disease reasoning: an open, literature-grounded knowledge graph and a five-task benchmark with a Graph-RAG retriever and reproducible harness. Our own audit shows that sentence-grounding and true literature weighting are necessary but not sufficient for a trustworthy graph. Precision remains modest even after we traced the original 13% fully-correct rate to concrete bugs—entity-disambiguation errors and a data-completeness issue—and fixed them; the corrected graph reaches only 45% correct-or-partial, because the bugs we found account for a minority of the extraction errors, with panel/list-co-occurrence artifacts, category errors, off-topic leakage, and lost relation valence still unaddressed (Appendix H). A compact 20–30-node, two-hop neighborhood is where Graph-RAG's own retrieval peaks, but a flat BM25 baseline over the same corpus reaches substantially higher gold-evidence coverage, so we do not yet have evidence that the graph structure itself outperforms simple passage retrieval.

We see AlzGraph less as a finished, validated resource than as a reproducible, repeatedly-audited starting point. The open questions we hope the community will pursue with the released harness follow directly from what we found: whether extraction precision can be raised substantially through a systematic medical-abbreviation resource, an explicit relation-valence field, and negation and list-co-occurrence filtering, without losing the sentence-grounded, true-paper-count properties that make the graph auditable in the first place; whether Graph-RAG's retrieval can be improved to close its coverage gap against flat lexical search; and how much genotype-dependent evidence actually improves anti-amyloid eligibility and safety decisions once that evidence is itself more reliable. By releasing the graph's true statistics, three rounds of audited extraction samples, and a harness that measures rather than assumes these effects, we hope to provide a foundation for studying knowledge-augmented reasoning in Alzheimer's disease honestly.

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Appendix Contents

A Related Work

Alzheimer's evidence base. AD is now defined biologically through the ATN biomarker framework [15], with plasma and imaging biomarkers such as p-tau217 [22] and blood-based markers more broadly [13] entering routine use alongside established genetic risk factors (*APOE* ε4 [8], *TREM2* [12], and autosomal-dominant genes tracked in longitudinal cohorts [2]) and a treatment landscape spanning long-standing cholinesterase inhibitors [4] and newer anti-amyloid antibodies [6, 23, 26]. This heterogeneity—genetics, fluid/imaging biomarkers, staging criteria, and a fast-moving treatment landscape with genotype-dependent safety constraints [10]—is exactly what motivates representing AD evidence as a typed, multi-hop graph rather than isolated facts.

Biomedical knowledge graphs. General-purpose biomedical KGs such as PrimeKG [7] and broader surveys of the space [9] provide wide disease coverage but are disease-agnostic by design: they do not encode the specific gene–biomarker–stage–treatment–outcome structure (e.g. genotype-dependent ARIA risk) that AD clinical reasoning requires. ALZKG takes the opposite trade-off, narrowing scope to one disease in exchange for a schema built around its actual clinical decision points.

Graph-RAG and knowledge-augmented LLMs. Retrieval-augmented generation grounds LLM outputs in external evidence [19], and graph-structured retrieval has been proposed to recover multi-hop reasoning chains that flat passage retrieval cannot [11]. LLMs have separately shown strong medical question-answering ability on general clinical knowledge [24]. ALZGRAPH's Graph-RAG retriever combines these lines directly: a personalized-PageRank walk over ALZKG serializes literature-weighted reasoning paths as LLM context (§4.1). Unlike more elaborate retrievers, it has no semantic/embedding-based fallback (Appendix D.1)—a deliberate scope choice we discuss as a limitation, not a claimed advantage.

Clinical NLP benchmarks and LLM evaluation. General medical QA benchmarks such as MedQA-USMLE [16] test broad clinical knowledge but not the integrative, multi-hop, safety-critical reasoning AD care specifically demands; ALZBENCH reuses a dementia-filtered subset of MedQA-USMLE as one externally sourced task (T4, §3.2) precisely so that at least one task is not self-authored by the model it evaluates. More broadly, evaluating LLMs on clinical tasks surfaces evaluation-methodology pitfalls of its own—silent scoring bugs, self-authorship confounds, item-pool scale—that we found and disclose directly rather than only report headline numbers (Appendix H).

B AlzKG Construction Details

B.1 Ontology Sources and Coverage

Table 6 lists every source integrated into the recognition lexicon, the layer(s) it feeds, and how many entities it contributes. Unlike a single-tier vocabulary, ALZKG draws on two tiers per layer where applicable: a small, hand-curated *seed* (81 entities total, §2.3) naming well-established AD concepts, and, for the gene layer specifically, a much larger *recognition catalog* (the complete HGNC approved protein-coding gene set) that is not curated to AD relevance at all (§2.3); only entities that survive the sentence-grounded relation filter (§2.3, Table 1) enter the released graph.

Table 6: Ontology and guideline sources used in ALZKG construction. "Seed" entities are hand-curated (`alzgraph/lexicon.py`); "Catalog" entities are harvested programmatically (`scripts/build_lexicon_from_ontologies.py`) and only enter the released graph if they clear the sentence-grounded relation filter. Access reflects each source's own public distribution terms as of this release; we report this descriptively rather than assert a specific license we have not independently verified.

Source	Layer(s)	Seed	Catalog	Access
NIA-AA ATN framework [15]	Biomarker, Stage	—	—	Published research framework
OMIM / ADSP / GWAS Catalog [3, 18]	Gene	25	—	Academic/open terms
HGNC	Gene	—	19,262	Public registry
MeSH	Biomarker, Stage	18 / 11	383 / 82	Public domain (NLM)
HPO [17]	Outcome	14	2,825	Open (publisher-stated)
ChEBI [14]	Treatment	—	5,702	Open (publisher-stated)
AAN / Alzheimer's Association [10]	Treatment	13	—	Published guideline
UMLS [5]	cross-ontology linking	—	$0^{\dagger}$	Free, registration req'd
Total		**81**	**28,254**	

†§2.5/Appendix B.5 discusses a lighter casefold-based substitute used in its place (merging entities that share a surface form) and a real bug this substitute introduced.

B.2 Entity Normalization Protocol

Recognition is a greedy, longest-n-gram token match (up to 8 tokens) over a compiled gazetteer, not a machine-learning NER model or an embedding-based fuzzy matcher: multi-word phrases take precedence over shorter substrings, gene symbols and other short, ambiguous strings are matched case-sensitively, and (post-fix, §2.5) any case-insensitive alias of 3 characters or fewer is dropped lexicon-wide. Surface forms are normalized to canonical entities through a curated synonym/abbreviation map (e.g. "Leqembi"→*Lecanemab*). Records from different ontology sources that share a surface form are merged into one canonical entity, but—since the extraction-precision audit found this merge rule itself could conflate unrelated concepts (§2.5)—only when the shared surface form is distinctive (multi-word or ≥5 characters) and the two records agree on layer. There is no semantic/embedding fallback for unresolved mentions: a genuine synonym or paraphrase absent from the curated alias map is simply not recognized, a recall gap distinct from (and not fixed by) the precision-focused audit and fix.

B.3 Relation Extraction Details

Every relation is rule-based: within a candidate sentence, a cross-layer entity pair induces a typed edge only if the sentence also contains one of that relation's curated trigger phrases (Table 7 lists a representative subset of each relation's full trigger set). ALZKG does not invoke an LLM-based extraction pass in this release (§2.3); all 5,065 released triplets are rule-based, a deliberate scope choice to avoid reporting unverified edges rather than a claim that rule-based extraction is more accurate (§2.5/Appendix B.5 shows it is not, by itself, reliable either).

Table 7: Relation-extraction trigger phrases (representative subset of each relation's full curated set; `scripts/build_kg_from_fulltext.py`). A relation fires only when a trigger substring co-occurs, in the same sentence, with a cross-layer entity pair of the listed layers.

Relation	**Layer pair**	**Trigger phrases (examples)**
risk_gene_for	gene → stage	risk, mutation, variant, carrier, predispos, susceptib
modulates_biomarker	gene → biomarker	modulat, regulat, increas, decreas, elevat, express
pharmacogenomic_consideration	gene → treatment	response to, genotype, contraindicat, tailor, metaboli
gene_associated_outcome	gene → outcome	associated with, predict, risk of, decline, progression
characterized_by_biomarker	stage → biomarker	characterized by, positive, elevated, hallmark, burden
biomarker_guides_treatment	biomarker → treatment	eligib, candidat, indicat, screen, stratif, positive
predicts_outcome	biomarker → outcome	predict, correlat, prognos, marker of, indicative of
treated_with	stage → treatment	treat, therap, first-line, approved, prescrib
associated_outcome	stage → outcome	associated with, progress, decline, lead to, convert
has_outcome	treatment → outcome	resulted in, led to, adverse, efficac, side effect

B.4 Knowledge Graph Statistics

Table 8 complements the summary in Table 1. Two structural properties are worth stating explicitly because they differ from a typical mined biomedical KG: every released triplet is cross-layer by construction (co-occurring same-layer entity pairs never induce an edge, §2.3), so there is no within-layer/cross-layer split to report; and every triplet is rule-based (§B.3), so there is likewise no rule-based/LLM-based split.

Table 8: Detailed ALZKG statistics (fully corrected graph, §2.5).

Statistic	**Value**	**(%)**
Total nodes	1,224	—
Total edges	5,065	100.0
Cross-layer edges	5,065	100.0
Rule-based edges	5,065	100.0
Median paper count	5	—
IQR paper count	3–10	—
Edges with paper count $\geq$10	1,314	25.9
Mean paper count	17.3	—

B.5 Extraction Precision Audit

§2.5 reports the headline numbers; this section gives the full protocol, both audits' judgments, and every fix in detail.

We drew a reproducible random sample of 30 triplets (seed 42) from an earlier version of the released graph (5,880 triplets, built from a then-current lexicon) and, for each, retrieved the actual corpus sentence used to emit it—the sentence whose recognized-entity list contains both the head and the tail—then judged whether that sentence supports the specific relation and direction claimed.

Of the 30: **4 (13%)** were fully correct, **8 (27%)** were partially or loosely supported (a real but broader, weaker, or reversed-direction association than the relation claimed), and **18 (60%)** were not supported by their own grounding sentence at all.

Diagnosing and fixing the entity-disambiguation cases. A recurring subset of the 18 unsupported cases were *entity-disambiguation collisions*: the gene symbol *NOS1* matched the unrelated clinical abbreviation "NOS" (Not Otherwise Specified); the gene *MET* and the compound L-lysine both matched one-letter amino-acid codes inside a protein-mutation notation ("Lys670_Met671"); the gene *IGFALS* matched the disease abbreviation ALS as a bare substring. Tracing these to their root cause in the entity-recognition code (`alzgraph/lexicon.py`, `scripts/build_lexicon_from_ontologies.py`) surfaced two genuine software bugs, not just curation gaps. *(1) Cross-concept alias merging:* the lexicon compiler merges two ontology records whenever they share any casefolded surface form, intended to catch the same real-world concept named twice across sources; but a short string like "MET"/"Met" collides by pure accident between the *MET* gene symbol and an amino-acid alias of methionine, and the merge silently fused them into one entity under the *gene* layer — so every case-insensitive mention of "met"/"methionine" anywhere in the 361,201-sentence corpus was being recognized as the *MET* gene. We fixed this by requiring the shared surface form to be distinctive (multi-word or $\geq$5 characters) and the two records' layers to match before merging. *(2) Order-dependent primary-symbol loss:* the raw HGNC gene table lists many genes' own official symbol as an *alias* of a different, unrelated gene (e.g. *NANOS1* lists "NOS1" as one of its aliases); our original conflict-resolution logic let whichever gene's record appeared first in the file claim a contested string, so *NOS1* itself silently lost recognition of its own primary symbol to *NANOS1* purely due to file order. We fixed this by reserving every gene's own approved symbol for itself before resolving alias-level conflicts. On top of these two bug fixes, we built a curated exclusion list (dictionary-word check via a standard word list, plus manually verified clinical-abbreviation, institutional-acronym, and domain-generic-term collisions) that drops a short gene alias — and, for a handful of audit-verified cases (*CBS*/corticobasal syndrome, *NPS*/neuropsychiatric symptoms, *PGC*/Psychiatric Genomics Consortium), even a gene's bare primary symbol — when it collides with a much more frequent non-gene meaning; *tau*/*TAU*, an official MAPT alias that this literature uses constantly as ordinary vocabulary for the tau protein rather than as a gene reference, is the largest single entry (removing it dropped *MAPT* out of the top-15 most-mentioned entities entirely). All three fixes are entirely in code (no hand-edits to specific triplets), so the resulting graph is rebuilt end-to-end from the same corpus and remains fully reproducible.

First repeat audit, and a second bug found while double-checking it. We rebuilt ALZKG with the fixed lexicon and drew a fresh n=30 sample (seed 42): precision rose to 6 (20%) fully correct, 12 (40%) partial, 12 (40%) unsupported. Before finalizing this number, we kicked off a from-scratch PMC re-fetch of the same 7,150 papers purely to double-check the faster in-place re-annotation used to produce that graph. The re-fetch instead surfaced an unrelated, previously-undisclosed bug: 943 of the 7,150 source papers (13%) had silently fallen back to abstract-only content in the original fetch (`source: "abstract"`, zero body sentences), and re-fetching the identical PMCIDs now successfully retrieves full text for nearly all of them — strong evidence this was a transient fetch failure during the original bulk run, not a stable "no full text available" property of those articles (which a fresh fetch would reproduce, not fix). This was large enough to matter: the recovered body text alone moved individual entity counts substantially (e.g. *MAPT*'s raw mention count differed by roughly 9$\times$ between the incomplete and complete corpus on an otherwise-identical lexicon), so we treat it as a second, independent bug fix rather than noise. We also discovered that the background re-fetch process itself had started before three of the lexicon fixes above were added and, because Python caches an imported module's state, had been silently annotating with a stale, partially-fixed lexicon for its entire run; we discarded its baked-in entity annotations and re-detected entities from its raw sentence text with the fully current code before rebuilding the graph a final time (5,065 triplets, Table 1).

Final repeat audit, pooled across two independent draws. We drew a fresh n=30 sample (seed 42) from this fully corrected graph: only 5 (17%) fully correct, 6 (20%) partial, 19 (63%) unsupported — worse than the interim result above, and worse than the original pre-fix audit's correct-or-partial rate. Rather than report this single, possibly-unlucky draw, we drew a second, independent n=30 sample (seed 43): 8 (27%) fully correct, 8 (27%) partial, 14 (47%) unsupported — a visibly different picture from the same graph. This 17%–40%-range spread across successive 30-item draws on graphs that differ only slightly is itself a finding: at n=30, a single audit is not a precise instrument for a graph this heterogeneous, and reporting whichever draw was run first (or looked best) would be misleading. We pool both seed-42-and-43 draws to n=60 as our final, reported estimate: **13 (21.7%) fully correct, 14 (23.3%) partial, 33 (55.0%) unsupported**. Correct-or-partial rises only from 40% (original) to 45% (pooled final) — a real but modest improvement, not the $\sim$20-point jump an earlier, less complete measurement of this same fix suggested. The pooled sample's failure modes: *gene-panel/co-occurrence-list artifacts* are the largest single new category — entities merely co-listed in a proteomics panel, a network-clustering analysis, a differential-diagnosis list, or a front-matter abbreviations glossary (e.g. a list defining "TLR2" and "AD" as separate entries) are extracted as if the paper's body had asserted a relation between them; this was present but less prominent in the smaller original sample. *Lab-technique-as-biomarker / methods-artifact category errors* recur heavily ("Blotting, Western" $\times$2, "Staining and Labeling", "Excluded" as a QC/eligibility term $\times$2, "Cell Culture Techniques"): an overly generic or mistyped MeSH/technique lexicon entry is recognized as a clinical biomarker or outcome. *Off-topic-disease leakage* persists at meaningful volume (ARDS/critical-care, perimenopause, and neuroleptic-malignant-syndrome sentences all contributed spurious AD triplets). *Directionality loss* remains common (galantamine shown reducing mortality, and gantenerumab shown slowing disease progression, are both still stored identically to a drug that causes the outcome). *Unhandled negation and hedging* persists too ("has not been linked to," "remains inconsistent"). None of these five failure modes is addressed by the lexicon fixes above; they are structurally different problems left as remediation directions (Appendix H). Full per-row judgments for all three audit rounds (original, first repeat, and both pooled final draws) are in `data/alzkg/extraction_audit.json`.

C AlzBench Dataset Details

C.1 Dataset Construction Protocols

T1 CDA. MCQ (n=20) and open-ended (n=8) items are hand-authored exam-style questions grounded in NIA-AA ATN staging, genetics, and treatment-safety knowledge (§3.2); MCQ distractors are clinically plausible but incorrect alternatives. **T2 CRG.** A local JSONL adapter mirrors the schema of a real memory-clinic report (patient history, cognitive assessment, biomarker panel → diagnostic impression); the public release ships one synthetic preview case rather than real ADNI/memory-clinic exports, which are access-restricted and not redistributed. **T3 BPM.** MCQs (n=10) are constructed from AAN/Alzheimer's Association appropriate-use-criteria rules pairing an APOE genotype and ATN biomarker profile with a correct anti-amyloid-antibody eligibility decision, including ARIA-risk contraindications. **T4 TR.** Items (n=16) are filtered from real MedQA-USMLE [16] using a word-boundary dementia/AD-term match, followed by a manual relevance pass (Appendix H); gold answers are MedQA-USMLE's own published answer key, not authored for this benchmark. Unlike some general-medicine QA benchmarks, no MMLU subset is included in this release. **T5 DRP.** Items (n=20) are randomly sampled real PMC abstracts from the AlzKG mining corpus (§2.3); there is no gold research plan to score against, so T5 is evaluated qualitatively on a subset (§4.5).

C.2 Gold Standard Annotation

No licensed clinician or domain-expert annotator was involved in constructing any AlzBench gold standard (Appendix H). T1/T3 gold answers were authored by the same LLM instance later evaluated on the Claude (direct) row of Table 5, a self-authorship confound discussed explicitly wherever that row is reported; T4's gold answers come from MedQA-USMLE's own external answer key rather than in-house authorship; T5 has no gold standard at all and is scored qualitatively. We report this plainly rather than describe an inter-annotator-agreement protocol we did not run.

C.3 Dataset Statistics

Table 9: ALZBENCH dataset statistics by task.

Task	Dataset	Instances	Gold standard	Split
T1 MCQ	Hand-authored	20	Hand-authored	Test only
T1 QA	Hand-authored	8	Hand-authored	Test only
T2 CRG	Synthetic preview	1	Synthetic	Test only (illustrative)
T3 BPM	AAN/AA-rule-derived	10	Rule-derived	Test only
T4 TR	MedQA-USMLE (dementia-filtered)	16	MedQA-USMLE answer key	Test only
T5 DRP	Real PMC abstracts	20 (2 used qualitatively)	None (qualitative)	Test only

D Graph-RAG Retriever Details

D.1 PPR Implementation

The retriever proceeds in three stages. First, the same dictionary-lexicon entity recognizer used in ALZKG construction (Appendix B.2) identifies seed entities in the query. Second, personalized PageRank is run from those seeds over the ALZKG adjacency (restart probability α=0.15), using a pure-Python implementation with an optional `networkx`-backed fast path, producing a relevance score for every reachable node. Third, the highest-scoring nodes (up to the node budget) are kept and all distinct source-to-sink reasoning paths within the depth budget are enumerated and truncated to the highest-scoring paths, each serialized as `(head, relation[N papers], tail)` chains annotated with relation type and literature weight. Unlike a hybrid retriever, there is no semantic/embedding-based fallback: a query whose entities are not recognized by the lexicon retrieves nothing, a scope limitation we do not currently address.

D.2 Hyperparameter Settings

Table 10 lists the retriever's hyperparameters and the values used to produce every result in this paper. Unlike a validation-tuned selection, the node-budget and depth defaults are *not* the settings that maximize gold-evidence coverage in §4.2's own sweep (peak coverage is at 20 nodes / 2 hops, Table 3); we keep the wider 30-node / 4-hop default to preserve fuller reasoning-path context for the LLM rather than to optimize this one intrinsic metric, and report the resulting trade-off explicitly rather than silently select the metric-maximizing setting.

Table 10: Graph-RAG retriever hyperparameters. "Swept" columns report the range tested in §4.2's ablation (Table 3); PPR restart probability and max serialized paths were not swept in this release.

Hyperparameter	**Value used**	**Range swept (Table 3)**
Max subgraph nodes	30	10, 20, 30, 40, 50
Max path depth	4	1, 2, 3, 4
PPR restart probability α	0.15	not swept
Max serialized paths	12	not swept

E Evaluation Metrics

All metrics below are pure-Python, standard-library implementations (`alzgraph/metrics.py`); this release does not depend on an external NLP package (e.g. the official ROUGE/SacreBLEU packages or a BERTScore/roberta-large embedding model) or a working LLM-as-judge (Appendix H), a deliberate no-external-dependency scope choice rather than an oversight.

Top-1 Accuracy. Exact match between the predicted option letter and the gold option letter, averaged over items.

ROUGE-L. Longest-common-subsequence F1 between whitespace-tokenized prediction and reference text.

Token-F1. Bag-of-tokens overlap F1 (precision and recall over the multiset of tokens) between prediction and reference.

BLEU-1. Clipped unigram precision between prediction and reference tokens (no brevity penalty).

Drug Safety Score. 1.0 if the selected option's text does not contain any contraindicated-agent substring for the given patient context (e.g. respecting ARIA exclusions), else 0.0.

Guideline Concordance. 1.0 if the selected option's text contains any guideline-recommended agent substring for the given context, else 0.0; defined in the released metrics code but not reported as a headline number in this paper's tables.

KG Evidence Coverage (KGEC). The fraction of the model's answer tokens that also appear among the tokens of the retrieved Graph-RAG paths—a lexical-overlap proxy for whether the answer actively used the retrieved evidence, not a correctness measure.

LLM-as-Judge. Part of the task design (§3.2) for open-ended CDA/CRG alignment and DRP scientific-validity/feasibility scoring, but no judge model, prompt, or agreement check exists in the released metrics code; no number in this paper uses it (Appendix H).

F Prompts

Unlike a single shared system prompt across tasks, each AlzBench task uses its own role-specific system prompt (reproduced verbatim below, `tasks/t*.py`); a task's user turn is the serialized Graph-RAG evidence (when the mode is `graph_rag`; a per-task literal prefix such as "AlzKG reasoning paths:" or "AlzKG context:") prepended to the task's own inputs (question/options, patient history, clinical scenario, or paper abstract, depending on the task).

T1 CDA — MCQ system prompt.

```
You are a cognitive neurologist taking an Alzheimer's disease
clinical decision exam.  Select exactly one option letter (A, B, C,
or D). Use NIA-AA biomarker (ATN) framing and guideline-consistent
reasoning.  Return only the option letter.
```

T1 CDA — open-ended QA system prompt.

```
You are a cognitive neurologist.  Answer the clinical question
in 2-4 concise sentences.  Name relevant disease stages, ATN
biomarkers (amyloid/tau/neurodegeneration), risk genes (e.g.  APOE),
treatments, contraindications, or outcomes when applicable.
```

T2 CRG system prompt.

```
You are a cognitive neurologist generating a diagnostic impression
for a memory-clinic report.  From the patient history, cognitive
assessment, and biomarker panel, produce an impression that
summarizes:  (1) the cognitive syndrome and severity, (2) the ATN
biomarker profile (amyloid, tau, neurodegeneration) and whether
it supports Alzheimer's disease, (3) the most likely diagnosis and
stage, and (4) management or follow-up recommendations.  Be concise,
evidence-grounded, and clinically safe.
```

T3 BPM system prompt.

```
You are a clinical neurologist specializing in Alzheimer's disease
pharmacotherapy.  Select the single most appropriate treatment from
A-D using APOE genotype, ATN biomarker status, and AAN / Alzheimer's
Association appropriate-use-criteria safety reasoning (including
ARIA risk).  Return only the option letter.
```

T4 TR system prompt.

```
You are a clinical neurologist managing Alzheimer's disease and
related dementias.  Select the safest guideline-consistent treatment
option from A-D, considering contraindications, ARIA risk for
anti-amyloid antibodies, comorbidities, and disease stage.  Return
only the option letter.
```

T5 DRP system prompt.

```
You are an Alzheimer's disease clinical researcher.  Given a paper
abstract, generate:  1.  a focused, novel research question, 2.  a
study design rationale (cohort, biomarkers, endpoints), 3.  the
required evidence or data and key feasibility considerations.  The
plan must be feasible, clinically meaningful, and grounded in known
gene-biomarker-stage-treatment-outcome evidence (ATN framework).
```

G Additional Experimental Results

Table 11 consolidates every original vs. final-corrected number in this paper (§2.5/Appendix B.5) in one place, for a reader who wants the full before/after picture without cross-referencing each subsection individually. "Original" is the graph as first released (5,880 triplets, pre-lexicon-fix, incomplete corpus); "Final" reflects both the lexicon fixes and the corpus-completeness fix (Appendix B.5). An interim, lexicon-fixed-but-not-yet-corpus-complete graph existed briefly during development and is not tabulated here, but is discussed in Appendix B.5 where it materially affected our own interpretation of the audit result (its correct-or-partial rate looked better than either the original or the final number).

Table 11: Every measured original vs. final number in this paper, in one place (§2.5, §4.3, §4.4). "—" marks a quantity not remeasured (Claude (direct), Appendix H) or unaffected by construction (BM25/No-RAG, which do not consult ALZKG's entity/edge content the way Graph-RAG and the KG-only baseline do).

Quantity	Task/setting	Original	Final
Entities	—	1,301	1,224
Triplets	—	5,880	5,065
Candidate sentences	—	361,201	320,299
Audit: fully correct (pooled)	n=60	4/30 (13%)*	13/60 (21.7%)
Audit: partial (pooled)	n=60	8/30 (27%)*	14/60 (23.3%)
Audit: unsupported (pooled)	n=60	18/30 (60%)*	33/60 (55.0%)
KG-only baseline accuracy	T1 CDA	0.55	0.50
KG-only baseline accuracy	T3 BPM	0.30	0.30
KG-only baseline accuracy	T4 TR	0.25 (chance)	0.375
KG-only accuracy (grounded)	T4 TR	0.23 (below chance)	0.42
Retrieval ablation gold coverage (peak)	node/depth sweep	0.10–0.17	0.10–0.17 (unchanged)
BM25 gold coverage	non-graph baseline	— (not run originally)	0.567

*The original audit was a single n=30 draw, not pooled; Appendix B.5 explains why we pool two draws for the final number.

H Limitations and Future Work

Item scale and self-authorship. T1's MCQ/QA and T3's items are curated, hand-authored exam-style questions (20, 8, and 10 items respectively after this revision's expansion from 10/3/5); T4 draws 16 items from real MedQA-USMLE via a dementia-term filter; T5 currently offers 20 real abstracts but only 2 have been used for the qualitative case study; T2 has exactly one item (§4.5) and is, in effect, unevaluated pending access to a non-restricted clinical-report dataset. These are still small samples for a benchmark claim. The self-authorship concern is specific to the *Claude (direct)* row: T1/T3 were authored by the same model instance that later answered them, which we believe explains its ceiling accuracy there more than any special capability of Graph-RAG. It does not apply to the three local open-source models in Table 5 (none of which authored any item), and their genuinely mixed, sub-ceiling results are correspondingly more informative, though still measured on the same small item pools. A more discriminating benchmark would (i) grow each item pool by an order of magnitude, (ii) source items independently of the evaluated model (as T4 already does), and (iii) deliberately include harder, rarer, or adversarially constructed cases (atypical presentations, conflicting guideline edge cases, recently updated appropriate-use criteria) where a model's parametric knowledge is less likely to already be saturated.

The Claude (direct) row is not rerun after either fix. Table 5's Claude (direct) row is reported as originally measured, on the original graph (§2.5), for both the No-RAG and Graph-RAG columns; it was not re-answered on either corrected graph the way the three open-weight rows were (§4.4). Every one of its conditions already sits at ceiling accuracy, attributed above to self-authorship (T1/T3) or same-model-family items (T4) rather than retrieval quality, so we judged a full manual re-answer unlikely to change the reported numbers and prioritized rerunning the open-weight rows instead, which are not ceiling-bound and are the table's more informative comparison. We flag this rather than silently present a mixed-vintage table as if every cell were measured under identical conditions.

Option-letter extraction from free-form text. Our answer-extraction routine originally case-folded a model's completion and returned the *first* standalone A/B/C/D match, which is fine for a short direct answer but silently wrong for a long chain-of-thought completion: the lowercase article "a" folds to "A" and can be matched well before a reasoning model states its actual conclusion. This first surfaced as an implausible Qwen3-8B T3 Graph-RAG accuracy built mostly out of such spurious matches; spot-checking the raw reasoning traces confirmed it (an early, uncorrected version of Table 5 reported T3 Graph-RAG 0.50 for Qwen3-8B). We fixed the extractor to prefer the *last* explicit "answer/option/choice is/might be X" phrase — robust to a model second-guessing itself mid-trace — falling back to the last case-sensitive bare letter only otherwise; every Qwen3-8B number in this paper uses the corrected extractor. Table 5's current T3 Graph-RAG value for Qwen3-8B (0.20, on the final, fully corrected graph) happens to numerically match the very first, pre-fix-extractor measurement on a since-superseded graph, which is a coincidence of three independent changes (the extractor fix and two rounds of graph fixes, §2.5) landing on the same value, not evidence that the extractor fix made no difference — the intermediate, lexicon-fixed-only graph measured 0.30 with the corrected extractor. We flag the original bug as a limitation because it is exactly the class of silent scoring bug that is easy to miss without inspecting raw model output, and a benchmark harness intended for others to trust should say so plainly rather than only report the corrected number.

Term-filtered external items are noisy. T4's dementia-term filter, even after switching to word-boundary matching (an early version matched *"mci"* inside *"triamcinolone"* and bare *"amyloid"* inside unrelated systemic-amyloidosis questions), still required a manual relevance pass that dropped 5 of 21 automatically filtered items because the matched term appeared only in a distractor option or a patient's incidental history rather than the tested clinical reasoning; scaling T4 will need a more precise filter or classifier rather than manual review.

Retriever determinism. An earlier version of the retriever iterated the graph's node set directly in several places (seed matching, PageRank state, and top-k neighborhood selection); because Python's `set` iteration order depends on a per-process string-hash seed, two runs of the identical query against the identical graph could silently keep a different tied path or seed order, occasionally changing the serialized Graph-RAG evidence (and, in one case we caught during evaluation, changing a downstream answer). We fixed this by iterating over a sorted node order everywhere ranking or state construction depends on it, and verified byte-identical output across five repeated runs with randomized hash seeds for both the retriever and the KG-only baseline; released code reflects this fix.

Corpus completeness, and a stale-import trap. While double-checking the extraction-precision fix (Appendix B.5) we found that 943 of 7,150 source papers (13%) had silently fetched abstract-only content in the original release rather than full text, almost certainly from transient PMC fetch failures rather than a genuine absence of full text, since re-fetching the identical papers now succeeds. We flag this as a limitation because it went undetected through the original release and every prior version of the audit: a per-paper fetch fallback that degrades gracefully (rather than erroring loudly) is exactly the kind of silent data-quality issue that is easy to miss without an independent re-fetch to compare against, and we do not have a systematic way to rule out further such gaps beyond the one we happened to find. Separately, and for the same reason worth naming explicitly: a background re-fetch process we launched to investigate this had itself started before three later lexicon fixes were added, and, because Python caches an imported module's state at first import, continued annotating with a stale, partially-fixed lexicon for its entire multi-hour run without any error — we caught this only by manually spot-checking one of its output entities against the lexicon then in effect. Both are now corrected in the released graph, but both illustrate that this pipeline's failure modes are not limited to the entity-disambiguation bugs that motivated the original audit.

Extraction precision (see also §2.5/Appendix B.5). Our original manual audit found only 13% of a random triplet sample fully correct and 60% unsupported by their own grounding sentence. We traced a recurring subset of this to two genuine software bugs (a cross-concept alias-merge that fused unrelated ontology entries sharing a short casefolded string, and an order-dependent conflict-resolution pass that could strip a gene's own primary symbol) plus a curatable set of short-symbol collisions, and separately found and fixed a data-completeness bug (13% of source papers had silently fetched abstract-only content). After fixing all of this, a pooled two-draw (n=60) re-audit found precision improved only modestly: 21.7% fully correct, 55.0% unsupported — correct-or-partial rises from 40% to 45%, not the larger jump an earlier, less complete measurement of this same fix suggested (Appendix B.5). The pooled re-audit's failure modes are mostly structurally different problems the fixes above do not address: gene-panel/co-occurrence-list artifacts (the single largest new category — entities merely co-listed in a panel, cluster, or glossary, extracted as if a relation were asserted between them) and lab-technique-as-biomarker category errors from overly generic or mistyped lexicon entries are now the dominant sources, alongside persisting directionality loss (no valence on the relation schema, so "improves" and "causes" are stored identically), off-topic-disease leakage, and unhandled sentence-level negation/hedging. Concrete remediation directions, none yet implemented in this release: extend the exclusion-list approach of §2.5 with a proper medical-abbreviation resource rather than a manually curated list, since residual collisions of that exact type should be expected (the repeat audits surfaced several more instances — IGF, PGC, TAU, and, in the final pooled sample, PVS/PVR and SCD — that earlier passes had not anticipated); require a candidate co-occurring entity pair to appear in ordinary prose rather than a list, panel, cluster, or glossary structure, since gene-panel/list artifacts are now the largest single failure mode; add a signed valence field to each relation rather than inferring it from the trigger phrase alone; add a lightweight negation/hedge detector (e.g. reject a sentence containing "no", "not", "unclear", or "remains unknown" within the trigger-phrase window); require the source sentence (or its paper) to also match an AD-anchored topic filter, not just contain a lexicon entity, to catch off-topic leakage that the audit still finds; and add allele-level entities (e.g., *APOE* $\varepsilon 2/\varepsilon 3/\varepsilon 4$) so genuinely protective and risk variants of the same gene are not collapsed onto one node, which is a real gap given the paper's own headline use case (genotype-dependent ARIA safety, §1) depends on exactly this distinction.

The non-graph retrieval comparison is intrinsic only. §4.2's BM25 comparison measures gold-token coverage, not downstream task accuracy: we did not re-run the full No-RAG/Graph-RAG language-model comparison (§4.4) under a third condition where models are given BM25-retrieved passages instead of Graph-RAG paths. Whether flat retrieval's higher coverage translates into better model-facing accuracy, worse accuracy (e.g., from noisier or less structured context), or no difference remains an open, measurable question with the harness as released.

No clinician review. No licensed clinician reviewed the hand-authored T1/T2/T3 items, the mined ALZKG triplets audited in §2.5, or any model output for clinical accuracy or safety. Every correctness judgment in this paper—item design, the extraction audit, drug-safety scoring—was made by an LLM or by the paper's authors, not by a domain expert blinded to the system under test. This matters more than usual here because the benchmark is explicitly framed around safety-critical decisions (ARIA eligibility, contraindications).

LLM-as-judge is unimplemented. §3.2/§3.3 describe an LLM-as-judge dimension for open-ended CDA, CRG, and DRP scientific-validity/feasibility scoring, but no judge model, prompt, or agreement check exists in the released metrics code, and no number in this paper uses it. All open-ended results reported here use only lexical overlap metrics (ROUGE-L, BLEU-1, Token-F1), which are known to correlate imperfectly with clinical correctness.

No societal-harm-specific safeguards beyond standard release practices. ALZKG and ALZBENCH are a research artifact for evaluating retrieval-augmented reasoning, not a clinical decision-support product; none of the released code or data should be used to make real treatment decisions without a licensed clinician, and the KG's coarse, sentence-level relation extraction can and does surface spurious associations (§2.3) that would need clinical review before any downstream use.

NeurIPS Paper Checklist

1. **Claims**
   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   Answer: [Yes]
   Justification: The abstract and §1 state exactly what is measured (mined KG statistics, three rounds of manual extraction-precision audit with the bugs found and fixed between them, an intrinsic retrieval analysis benchmarked against a non-graph baseline, a deterministic KG-only baseline, and a blind No-RAG vs. Graph-RAG comparison run against four models: Claude direct and three local open-weight models), including the unflattering findings (13%→21.7% fully-correct audited-triplet precision, i.e. 45% correct-or-partial, still well short of reliable; BM25 exceeding Graph-RAG's own coverage metric), rather than implying either a more reliable graph or a broader model comparison than we ran.

2. **Limitations**
   Question: Does the paper discuss the limitations of the work?
   Answer: [Yes]
   Justification: Appendix H discusses item-pool scale (including T2's unevaluated, n=1 status), self-authorship of the T1/T3 items, T4's term-filter noise, a retriever-determinism bug we found and fixed, the extraction-precision audit's findings and remediation directions, the intrinsic-only scope of the non-graph retrieval comparison, the absence of any clinician review, and the unimplemented LLM-as-judge dimension.

3. **Theory Assumptions and Proofs**
   Question: Does the paper include theoretical results requiring stated assumptions and proofs?
   Answer: [NA]
   Justification: The paper is empirical/systems-oriented (a knowledge graph, a benchmark, and a retriever); it contains no theorems.

4. **Experimental Result Reproducibility**
   Question: Does the paper disclose all information needed to reproduce the main experimental results?
   Answer: [Yes]
   Justification: The KG-only baseline is deterministic and reproducible with a single command per dataset (§4.3); the Claude-direct row's exact prompts, queue, and cache format are described in §4.4 and implemented in the released evaluation code; the extraction-precision audit's triplet samples (both pooled seeds) are reproducible with a fixed random seed each and a released script, and the non-graph retrieval baseline is a released, deterministic script over the same corpus and probes as the Graph-RAG ablation; the scripts used to construct each task's dataset are released alongside the data they produced.

5. **Open Access to Data and Code**
   Question: Does the paper provide open access to the data and code, with sufficient instructions?
   Answer: [Yes]
   Justification: Code and data are included as supplementary material for review and will be published at a public repository on acceptance (identity withheld for anonymity per Appendix H); a manifest included in the release maps every paper component to its release code.

6. **Experimental Setting/Details**
   Question: Does the paper specify all training/evaluation details needed to understand the results?
   Answer: [Yes]
   Justification: §4.1 (Setup) gives PPR hyperparameters, neighborhood/path budgets, and prompting protocol; §4.3–4.4 give the exact evaluation commands and item counts per task.

7. **Experiment Statistical Significance**
   Question: Does the paper report error bars or other measures of statistical significance?
   Answer: [No]
   Justification: Item counts (8–20 per task) are too small for meaningful confidence intervals on a single-model, single-run comparison; we report raw accuracy/counts throughout and flag small-n explicitly rather than presenting unsupported error bars (Appendix H).

8. **Experiments Compute Resources**
   Question: Does the paper provide sufficient information on the computer resources needed?
   Answer: [Yes]
   Justification: KG mining, PPR retrieval, the KG-only baseline, and the Claude-direct evaluation run on CPU in well under an hour on a single machine, with no distributed training involved (§4.2 reports mean per-query retrieval latency, 30–70 ms). The three local-model rows in Table 5 (§4.4) additionally used one GPU (a single NVIDIA H200 or RTX 6000, model-dependent, of a shared multi-GPU machine) via a local Ollama server, with each model's full 8-condition run (4 tasks ×

2 modes) completing in a few minutes to under an hour depending on model size and whether the model's chain-of-thought mode was enabled.

9. **Code Of Ethics**
   Question: Does the research conform to the NeurIPS Code of Ethics?
   Answer: [Yes]

10. **Broader Impacts**
    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work?
    Answer: [Yes]
    Justification: See §*Broader Impact* above.

11. **Safeguards**
    Question: Does the paper describe safeguards for responsible release of models/data with a high risk of misuse?
    Answer: [NA]
    Justification: The release is a knowledge graph mined from public open-access literature and a benchmark harness, not a generative model or a dataset of sensitive personal data (the private ADNI/memory-clinic-derived T2 adapter is explicitly not redistributed, §3.2); we do not see a specific high-risk-of-misuse vector beyond those already discussed in §*Broader Impact*.

12. **Licenses**
    Question: For assets used, are the license and terms of use respected and properly cited?
    Answer: [Yes]
    Justification: Ontology/guideline resources (NIA-AA, OMIM, GWAS Catalog, MeSH, HPO, ChEBI, UMLS, AAN/AA) and the MedQA-USMLE dataset [16] are cited at first use (§2, §2.2); the release's own license is Apache-2.0.

13. **Assets**
    Question: Are new assets introduced in the paper well documented?
    Answer: [Yes]
    Justification: ALZKG, ALZBENCH, and the Graph-RAG retriever are documented in §2–3 and in the accompanying documentation released with the code, including provenance/honesty notes on how each artifact was produced.

14. **Crowdsourcing and Research with Human Subjects**
    Question: Does the paper involve crowdsourcing or research with human subjects?
    Answer: [NA]
    Justification: No human-subjects data collection was performed; all clinical text is either literature (PMC open-access articles) or de-identified, publicly released benchmark items (MedQA-USMLE) or curated exam-style items.

15. **IRB Approvals**
    Question: Does the paper describe potential risks to participants and IRB approval, if applicable?
    Answer: [NA]
    Justification: No human subjects were involved (see previous item); T2's private ADNI/memory-clinic adapter operates only on data the institution already holds under its own IRB-approved protocols and is not redistributed by this release.

16. **Declaration of LLM Usage**
    Question: Does the paper describe the usage of LLMs in the core methodology of the research?
    Answer: [Yes]
    Justification: LLMs are a core, disclosed part of the methodology in §4.4: Claude (Sonnet 5) both authored some T1/T3 benchmark items and, separately, served as the evaluated model for the Claude-direct row, blind to gold labels during the latter role; three open-weight local models (Llama-3.2-3B, Gemma3-4B, Qwen3-8B) additionally served as evaluated models proper, with no authorship role. These roles and their implications for the reported ceiling accuracy (Claude) versus genuinely mixed accuracy (local models) are discussed explicitly in §4.4 and Appendix H rather than left implicit.